南開大學 作业纸

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习题7.1

$\boxed{A组}$

6. $|\vec{a}| = \sqrt{6^2+7^2+6^2}$

$= 11$

$\therefore \frac{\vec{a}}{|\vec{a}|} = \left\{\frac{6}{11}, \frac{7}{11}, -\frac{6}{11}\right\}$

7. $A(1,2,3)$, $B(3,-4,6)$

$\vec{AB} = (2, -6, 3)$

$|\vec{AB}| = \sqrt{2^2+(-6)^2+3^2}$

$= \sqrt{4+9+36} = 7.$

$\cos\alpha = \frac{x}{\sqrt{x^2+y^2+z^2}} = \frac{2}{7}$

$\cos\beta = \frac{y}{\sqrt{x^2+y^2+z^2}} = -\frac{6}{7}$

$\cos\gamma = \frac{z}{\sqrt{x^2+y^2+z^2}} = \frac{3}{7}$

12. $(\vec{a}+\vec{b}+\vec{c})^2$

$= (\vec{a}+\vec{b}+\vec{c})(\vec{a}+\vec{b}+\vec{c})$

$= \vec{a}^2+\vec{b}^2+\vec{c}^2 + 2\vec{a}\cdot\vec{b} + 2\vec{b}\cdot\vec{c} + 2\vec{a}\cdot\vec{c}$

$= 4^2+2^2+6^2 + 2\times4\times2\times\cos\frac{\pi}{3} + 2\times6\times2\times\cos\frac{\pi}{3}$

$+ 2\times6\times4\times\cos\frac{\pi}{3}$

$= 56+8+12+24 = 100$

$\Rightarrow |\vec{a}+\vec{b}+\vec{c}| = \sqrt{100} = 10.$

14. $\lambda\vec{a}+\mu\vec{b}$

$= (3\lambda, 5\lambda, -2\lambda) + (2\mu, \mu, 4\mu)$

$= (3\lambda+2\mu, 5\lambda+\mu, -2\lambda+4\mu)$

若使 $\lambda\vec{a}+\mu\vec{b}$ 与 z 轴垂直，则

$z=0$

$-2\lambda+4\mu=0 \Rightarrow \lambda=2\mu.$

17. $\vec{OA} = \vec{i}+3\vec{k}$, $\vec{OB} = \vec{j}+3\vec{k}.$

$S_{\triangle OAB} = \frac{1}{2}|\vec{OA}\times\vec{OB}|$

$\vec{OA}\times\vec{OB} = \begin{vmatrix} i & j & k \\ 1 & 0 & 3 \\ 0 & 1 & 3 \end{vmatrix} = (-3, -3, 1)$

$\therefore |\vec{OA}\times\vec{OB}| = \sqrt{3^2+3^2+1} = \sqrt{19}$

$\therefore S_{\triangle OAB} = \frac{1}{2}\sqrt{19}.$

$\boxed{B组}$

2. 设 $\vec{c}$ 与 $\vec{b}$ 的角平分线为 $\vec{a}$.

则 $\widehat{\vec{c}\,\vec{a}} = \widehat{\vec{b}\,\vec{a}}$，夹角相等.

$\cos\langle\vec{a},\vec{c}\rangle = \cos\langle\vec{b},\vec{a}\rangle.$

设 $\vec{a} = (x, y, z)$

$\Rightarrow \cos\langle\vec{a},\vec{c}\rangle = \frac{7x-4y-4z}{9\times\sqrt{x^2+y^2+z^2}}$

$\cos\langle\vec{b},\vec{a}\rangle = \frac{-2x-y+2z}{3\sqrt{x^2+y^2+z^2}}.$

两者相等，则

$7x-4y-4z = -6x-3y+6z$

$13x-y = 10z.$

$\because |\vec{b}| = \frac{1}{3}|\vec{c}|.$

$\Rightarrow 3\vec{b}+\vec{c}$ 与 $\vec{a}$ 平行

$\Rightarrow (1,-7,2) \parallel \vec{a}$

$\Rightarrow |\vec{a}| = 5\sqrt{6}$

设 $\vec{a} = (x, -7x, 2x)$

$\sqrt{54}x = 5\sqrt{6}$

$\Rightarrow x = \frac{5}{3}$

$\therefore \vec{a} = \left(\frac{5}{3}, -\frac{35}{3}, \frac{10}{3}\right)$

习题7.2

$\boxed{\text{A类}}$

1. $\vec{n}=(6,-4,7)$ 即为平面法向量

则 $6x-4y+7z+m=0$

代入 $(2,3,-1)$ 得 $12-12-7+m=0$

$\Rightarrow m=7$

$\therefore$ 平面方程为 $6x-4y+7z+7=0$

4. 设此平面法向量坐标为 $\vec{n}=(a,b,c)$，则：

$\vec{n}$ 与两个平面法向量都垂直

$\vec{n_1}=(3,-2,2)$，$\vec{n_2}=(5,-4,3)$

$$\begin{cases}\vec{n}\cdot\vec{n_1}=0\\ \vec{n}\cdot\vec{n_2}=0\end{cases} \Rightarrow \begin{cases}3a-2b+2c=0\\ 5a-4b+3c=0\end{cases} \Rightarrow \begin{cases}a=-c\\ b=-\frac{1}{2}c\\ c=c\end{cases}$$

$\therefore$ 平面的法向量为 $\vec{n}=(-1,-\frac{1}{2},1)$

$\therefore -x-\frac{1}{2}y+z=m$

$\Rightarrow$ 代入 $(3,-1,-5)$

$-3+\frac{1}{2}-5=m \Rightarrow m=-\frac{15}{2}$

$\therefore -x-\frac{1}{2}y+z+\frac{15}{2}=0$

$\Rightarrow x+\frac{1}{2}y-z-\frac{15}{2}=0$

平面方程为 $2x+y-2z+15=0$

5(1) $\begin{cases}x-y+z+5=0\\ 3x-8y+4z+36=0\end{cases}$

令 $x=0$，代入 $\begin{cases}z-y+5=0\\ -8y+4z+36=0\end{cases}$

$\Rightarrow \begin{cases}y=4\\ z=-1\end{cases}$

$\therefore$ 直线上一点为 $(0,4,-1)$

方向向量为 $(1,-1,1)\times(3,-8,4)$

$$=\begin{pmatrix} i & j & k\\ 1 & -1 & 1\\ 3 & -8 & 4\end{pmatrix}=\{4,-1,-5\}$$

$\Rightarrow$ 标准方程

$$\frac{x}{4}=\frac{y-4}{7}=\frac{z+1}{-5}$$

$\Rightarrow$ 参数方程为

$$\begin{cases}x=4t\\ y=-t+4\\ z=-5t-1\end{cases}$$

9. 直线 L_1 的方向向量为

$\vec{s_1}=\vec{n_1}\times\vec{n_2}=(1,2,1)\times(1,-2,1)$

$$=\begin{pmatrix} i & j & k\\ 1 & 2 & 1\\ 1 & -2 & 1\end{pmatrix}=(4,0,-4)$$

$\vec{s_2}=\vec{n_3}\times\vec{n_4}=(1,-1,-1)\times(1,-1,2)$

$$=\begin{pmatrix} i & j & k\\ 1 & -1 & -1\\ 1 & -1 & 2\end{pmatrix}=(-3,-3,0)$$

$$\cos\theta=\frac{|-12+0+0|}{4\sqrt{2}\times 3\sqrt{2}}=\frac{1}{2}$$

$\therefore \theta=\frac{\pi}{3}$

12. $\begin{cases}x=t+1\\ y=2t-1\\ z=t\end{cases}$ $\begin{cases}x=t+2\\ y=2t-1\\ z=t+1\end{cases}$

L_1: $\frac{x-1}{1}=\frac{y+1}{2}=\frac{z}{1}$

L_2: $\frac{x-2}{1}=\frac{y+1}{2}=\frac{z-1}{1}$

两直线平行，设 $P(t_1+1,2t_1-1,t_1)$

$Q(t_2+2,2t_2-1,t_2+1)$

$|PQ|^2=(t_2-t_1+1)^2+(2t_2-2t_1)^2+(t_2-t_1+1)^2$

$=2(t_2-t_1+1)^2+4(t_2-t_1)^2$

$\Rightarrow$ 令 $t=t_2-t_1$，则

$|PQ|^2=2(t+1)^2+4t^2=6t^2+4t+2$

$=6(t^2+\frac{2}{3}t+\frac{1}{9})+\frac{12}{9}\geq\frac{12}{9}$

$\therefore |PQ|\geq\frac{2\sqrt{3}}{3}$

$\therefore$ 两直线之间的距离为 $\frac{2}{3}\sqrt{3}$

习题7.3

$\boxed{\text{A组}}$

3. $\begin{cases}(x-4)^2+(y-7)^2+(z+1)^2=36\\ 3x+y-z=9\end{cases}$

$\Rightarrow (x-4)^2+(y-7)^2+(z+1)^2=12x+4y-4z$

$\Rightarrow x^2-8x+16+y^2-14y+49+z^2+2z+1=12x+4y-4z$

$x^2-20x+y^2-18y+z^2+6z+66=0$

$\Rightarrow (x-1)^2+(y-6)^2+z^2=25.$

$\Rightarrow$ 圆心 $(1,6,0)$, $r=\sqrt{25}=5.$

6. $\begin{cases}y=z^2+1\\ x=0.\end{cases}$

绕y轴 $\Rightarrow f(y,\pm\sqrt{x^2+z^2})=0$

$\Rightarrow y=x^2+z^2+1$

绕z轴 $\Rightarrow f(\pm\sqrt{x^2+y^2},z)=0$

$\Rightarrow z=x^2+y^2+1$

绕y轴和z轴

$\Rightarrow z^2+1=\pm\sqrt{x^2+y^2}$

$\Rightarrow z^4+2z^2+1=x^2+y^2.$

讨论为 $x^2+y^2-z^4-2z^2-1=0$

11. 曲线 $\begin{cases}x^2+y^2+z^2=R^2\\ x^2+y^2=z^2,\end{cases}$

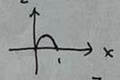
代入得 $\begin{cases}2x^2+2y^2-R^2=0\\ z=0\end{cases}$

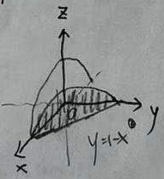
$\Rightarrow$ 以原点为圆心，$\frac{R}{\sqrt{2}}$ 为半径的一个圆

$\boxed{\text{B组}}$

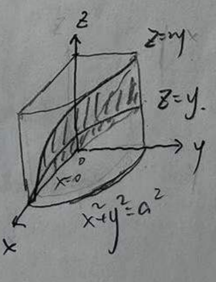
4. (1) $\begin{cases}z=1-x^2-y^2\\ y=1-x\\ y=0\\ z=0\end{cases}$

代入得 $z=1-x^2-(1-x)^2=1-x^2-1-x^2+2x=-2x^2+2x$





(2) $\begin{cases}z=y\\ z=2y\\ x^2+y^2=a^2\\ x=0\end{cases}$



习题8.1

7(1) $\lim\limits_{\substack{x\to0\\y\to0}} \dfrac{x^2+y^2}{x^2+y^2+(x-y)^2}$

用 $x=0$ 趋近 得原式 $= \dfrac{y^2}{2y^2} = \dfrac{1}{2}$

用 $y=mx$ 趋近得 $(m\neq0)$

原式 $= \lim\limits_{x\to0} \dfrac{x^2+m^2x^2}{m^2x^2+x^2+(x-mx)^2}$

$= \dfrac{m^2+1}{m^2+1+(1-m)^2} = \dfrac{m^2+1}{2m^2+2-2m} \neq \dfrac{1}{2}$

两者极限不同，则原式极限不存在.

8(1) $\lim\limits_{\substack{x\to0\\y\to0}} \dfrac{e^x+e^y}{\cos x+\sin y}$

原极为二元初等函数，则

$\lim\limits_{\substack{x\to0\\y\to0}} f(x,y) = f(0,0) = \dfrac{e^0+e^0}{\cos 0+\sin 0} = 2$.

8(4) $\lim\limits_{\substack{x\to\infty\\y\to a}} (1+\dfrac{1}{xy})^{\frac{x^2}{x+y}}$

$= \lim\limits_{\substack{x\to\infty\\y\to a}} \left[(1+\dfrac{1}{xy})^{xy}\right]^{\frac{x}{y(x+y)}}$

$= e^{\lim\limits_{\substack{x\to\infty\\y\to a}} \frac{x}{y(x+y)}} = e^{\frac{1}{a}}$.

9(1) $f(x,y) = \dfrac{x^2+y^2}{x^2+y^4}$，$a=\infty, b=\infty$.

$\lim\limits_{x\to a}\lim\limits_{y\to b} f(x,y) = \lim\limits_{x\to a}\left[\lim\limits_{y\to\infty} \dfrac{x^2+y^2}{x^2+y^4}\right] = 0$

$\lim\limits_{y\to b}\lim\limits_{x\to a} f(x,y) = \lim\limits_{y\to b}\left[\lim\limits_{x\to\infty} \dfrac{x^2+y^2}{x^2+y^4}\right] = 1$

10. 证明：

$$f(x,y) = \begin{cases} \dfrac{\ln(1+xy)}{x}, & x\neq0 \\ y, & x=0 \end{cases}$$

定义域 $xy>-1$

① 当 $(0,0)$ 处，当 $x\neq0$ 时，

$f(x,y) = \dfrac{\ln(xy+1)}{x} = y\dfrac{\ln(xy+1)}{xy} = y\ln(xy+1)^{\frac{1}{xy}}$，$y\neq0$

$f(x,y)=0$，$y=0$

$\because \lim\limits_{\substack{x\to0\\y\to0}} \ln(1+xy)^{\frac{1}{xy}} = 1 \Rightarrow \left|\ln(1+xy)^{\frac{1}{xy}}\right| < 2$

取 $\delta = \dfrac{\varepsilon}{2}$，则

当 $0<|x|,|y|<\delta$ 时，原 $= \left|y\ln(1+xy)^{\frac{1}{xy}}\right| \leq 2|y| < \varepsilon$

$\therefore \lim\limits_{\substack{x\to0\\y\to0}} f(x,y) = 0 = f(0,0)$

② 在 $(0,\bar{y})$ 处，$\bar{y}\neq0$

当 $x\neq0$ 时，

$|f(x,y) - f(0,\bar{y})| = |y\ln(1+xy)^{\frac{1}{xy}} - 1 + y - \bar{y}|$

$\leq |y|\,|\ln(1+xy)^{\frac{1}{xy}} - 1| + |y-\bar{y}|$

当 $x=0$ 时，

$|f(x,y) - f(0,\bar{y})| = |y-\bar{y}|$

$\lim\limits_{\substack{x\to0\\y\to\bar{y}}} \ln(1+xy)^{\frac{1}{xy}} = 1$，$\bar{y}\neq0$ 时.

$\therefore f(x,y)$ 也在 $(0,\bar{y})$ 处连续

综上所述，$f(x,y)$ 在定义域上连续